Learning-Based Contour Selection for Boundary Refinement in Diffusion-Based Monocular Depth Estimation

Anonymous Author(s)
Affiliation withheld for blind review

***Abstract*—Monocular depth estimation based on diffusion models achieves high accuracy on smooth surfaces but tends to blur depth discontinuities (object boundaries). An approach to address this issue is to sharpen the predicted depth along contours detected by a segmentation model. However, applying sharpening indiscriminately to all contours degrades the boundary F1 score. In this study, we observe that this degradation stems from sharpening contours that lack actual depth steps. To address this, while keeping the sharpening operation fixed, we employ a learned selector to identify only the relevant contours for sharpening. Experiments on real-world data demonstrate that the proposed method improves the boundary F1 score by 11.9% on NYUv2. Furthermore, applying the same selector without retraining to a depth model with a different architecture and output representation achieves a 27.8% improvement.**

***Index Terms*—monocular depth estimation, depth boundaries, segmentation, diffusion models**

I. Introduction

In recent years, significant progress has been made in applying the prior knowledge of large-scale image generation models to monocular depth estimation. For instance, Marigold [1] fine-tunes the diffusion-based image generation model Stable Diffusion on synthetic RGB-D data, achieving high generalization performance with a small amount of training data. Lotus [2] builds upon this by replacing the multi-stage diffusion process with a single-step prediction, significantly accelerating inference while maintaining accuracy. Furthermore, Marigold V2 [3] improves accuracy by utilizing a diffusion transformer originally designed for image editing.

Despite their high overall accuracy, these methods face a common challenge: blurring depth discontinuities at object boundaries. This issue is believed to stem from the generative model's inherent prior for smoothness, as well as its architectural nature of operating in a low-resolution latent space, which causes depth steps at occlusion boundaries to spread over several pixels. Because boundary pixels constitute only a tiny fraction of the image, this degradation is difficult to capture using standard image-wide evaluation metrics; nevertheless, it becomes clearly apparent when evaluated using metrics specifically designed for object boundaries [5].

To address the blurring of depth discontinuities, methods have been proposed that improve boundaries through post-processing. These methods do not require modifications to the depth model and can be applied to the output of any model. Previous research has focused on learning how to correct depth, including methods such as learning displacement fields that reposition pixels around occlusion boundaries [7] and post-processing that uses edges as cues [9].

A simpler approach involves using segmentation results as a guide. Segmentation models such as Segment Anything [11] can segment an image into objects and regions and extract their contours with high accuracy. Object contours frequently correspond to locations of abrupt depth changes. Therefore, we refill the predicted depth on both sides of the contour to make blurry steps more abrupt (hereafter referred to as "sharpening"). However, when sharpening is applied to all contours in Marigold V2, the boundary BF_1 actually decreases. This is because the majority of segmentation contours—such as wallpaper patterns or floor seams—are contours that do not involve depth discontinuities. Applying sharpening to such contours introduces discontinuities that do not actually exist.

These observations suggest that the effectiveness of sharpening depends far more on which contours are selected for processing than on adjusting their exact spatial positions. Indeed, when ground-truth depth steps are used to selectively filter the contours—without altering their original positions at all—depth estimation performance improves.

Inspired by this insight, we propose a learning-based contour selection framework to selectively identify depth-discontinuous boundaries. In our approach, the underlying depth estimation model and segmentation model remain entirely fixed. Instead, we introduce a lightweight U-Net as a binary selector, which takes the RGB image, the predicted depth, and the detected contours as inputs. Because only this small-scale selector is trained, any performance improvement can be strictly attributed to its ability to identify appropriate contours for sharpening.

This study makes the following three main contributions:

1) **Learning-based contour selection:** We propose a post-processing framework that retains a fixed sharpening operation while introducing a learned selector to identify the optimal contours for depth refinement.
2) **Superior boundary accuracy:** We demonstrate that our proposed method outperforms both non-learning-based baselines and classical edge-preserving filters, achieving significant improvements in boundary accuracy.
3) **Transferability and dataset sensitivity:** We show that while the trained selector generalizes well across different depth model architectures without retraining, its

performance is sensitive to dataset shifts, highlighting key requirements for cross-domain transferability.

II. Related Work

A. Diffusion-based monocular depth estimation

Marigold [1] fine-tunes the image diffusion model Stable Diffusion on synthetic RGB-D data to estimate affine-invariant depth. Lotus [2] predicts the depth itself rather than noise and replaces the multi-step process with a single step at a fixed timestep, improving both accuracy and speed. Garcia et al. [4] showed that correcting Marigold's inference procedure and combining single-step inference with end-to-end fine-tuning makes it orders of magnitude faster without loss of accuracy. Marigold V2 [3] builds on a diffusion transformer for image editing and estimates affine-invariant log depth in a single step. All of these improve the depth model itself; this work leaves the model untouched and post-processes its output.

B. Improving depth boundaries

Ramamonjisoa et al. [7] learn a 2D displacement field that re-samples pixels around occlusion boundaries, improving the output of any depth estimator by post-processing. It uses NYUv2 with the same 795 training and 654 test frames and is the closest prior work. SharpNet [8] builds occluding-contour estimation into the training of the depth network, and Peng et al. [9] reduce the degradation near occlusion boundaries in self-supervised monocular depth with an edge-guided post-processing step. Muthukkumar [10] fuses a monocular estimate with depth from a neural radiance field to recover detail. All of these design or learn how to correct the depth; this work fixes the correction and learns only where to apply it. Since no trained weights are released for [7], we do not compare against it directly.

C. Segmentation and depth estimation

Segment Anything [11] is a segmentation model that extracts arbitrary regions of an image without being told the object class. This work uses its contours as candidates for depth boundaries and learns to select those that carry a depth step. To our knowledge, no prior work uses the contours of a segmentation model as candidates for depth boundaries.

D. Boundary metrics

Pixel-averaged metrics such as AbsRel, the mean absolute error relative to the ground-truth depth, and δ_1, the share of pixels whose prediction is within a factor of 1.25 of ground truth, hardly reflect blurred boundaries, since boundary pixels are a small fraction of the image. Depth Pro [5] proposed a scale-invariant boundary metric, BF_1, that declares a depth step where the ratio of neighbouring depths exceeds a threshold, sweeps that threshold from 5% to 25%, and scores the agreement with F1. SMD-Nets [6] proposed the Soft Edge Error (SEE) for evaluating stereo, measuring the error at ground-truth boundary pixels while tolerating small displacements, and Marigold V2 [3] also uses it. We use both and propose no new metric.

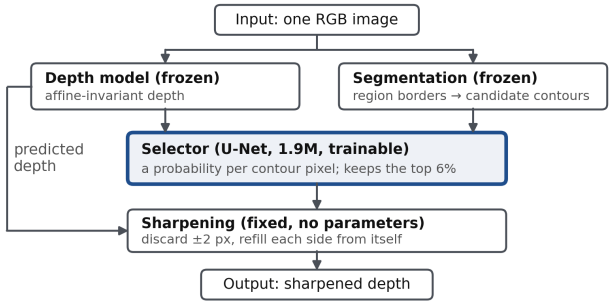


Fig. 1. Pipeline of the method. From the input RGB image, the fixed depth model predicts depth and the fixed segmentation model extracts candidate contours, and sharpening is applied only at the contours the selector keeps. Only the selector is trained.

III. Method

A. Overview

Fig. 1 gives an overview of the method. The input is a single RGB image. The depth model predicts affine-invariant depth, and the segmentation model extracts candidate contours from region borders. The selector takes the image, the predicted depth and the contours, outputs for each contour pixel the probability that it lies on a true depth step, and keeps a fixed fraction of the most probable. Finally, the barrier sharpening of Sec. III-B is applied at the kept contours. Only the selector is trained: the depth model and the segmentation model are fixed, and barrier sharpening has no learned parameters.

B. Barrier sharpening

Diffusion-based depth estimators mix foreground and background depth at object boundaries, turning a step that should change within one pixel into a slope over several. If the boundary is known, the step can be restored by discarding the mixed pixels around it and refilling each side from its own depth alone. We take the boundary from the contours of a segmentation model and, because each contour acts as a barrier to interpolation, call this operation *barrier sharpening*.

Let z be a predicted depth map, V its valid pixels and $C \subseteq V$ a set of contour pixels. The operation discards the band

$$B = \{\, p \in V \mid \mathrm{dist}(p, C) \le r_b \,\}, \tag{1}$$

labels the connected components of $V \setminus B$, so that the two sides of a contour receive different labels, and floods those labels inward one pixel at a time until B is covered, giving every $p \in B$ a label $\ell(p)$. Each discarded pixel is then refilled only from pixels outside the band that carry its label, that is, from its own side of the contour,

$$\hat{z}(p) = \frac{1}{|N(p)|} \sum_{q \in N(p)} z(q), \tag{2}$$

where $N(p) = \{\, q \notin B \mid \|q - p\| \le r_f,\ \ell(q) = \ell(p) \,\}$. Pixels outside B are unchanged. The component labels come from the contours themselves, not from the segmentation masks. The operation has only two settings, the band width r_b and the refill radius r_f, and learns nothing.

The separation of the two sides holds only where a contour closes or reaches the image border. Where it ends inside the image the valid region stays connected around that end, so both sides of the band belong to one component there and one-sided interpolation does not hold. Selection makes this more common, since keeping only part of the contour pixels leaves fragments rather than whole outlines. Both effects weaken the repair, and both are present in every number we report.

C. The selector

The selector is a network that keeps, among the contours extracted by the segmentation model, only those that carry a depth step. It is trained as a binary classifier that decides for each contour pixel whether it lies on a true depth step, and at inference only the most probable contour pixels are passed to barrier sharpening.

Inputs and output. The selector is a U-Net that takes the RGB image (3 channels), the predicted depth (1) and the contour mask (1), and emits one logit per pixel. The prediction is standardised per image. For a model that emits disparity v rather than log depth we apply $-\log\max(v, 10^{-3})$ first: disparity rises as depth falls, so feeding it raw would invert the signal, and an affine-invariant disparity can be non-positive, which the clamp handles. This conversion is only an approximation, and the Lotus results in Table II include its effect.

Retention. Rather than thresholding the probability, we keep a fixed fraction ρ of the contour pixels per image, in order of probability. A fixed threshold keeps a different amount depending on how many contours a scene happens to contain, so the budget the repair spends would drift from frame to frame. ρ is set to the share of positive contour pixels in the training data.

Labels and loss. A contour pixel is positive if it lies within 1 px of a depth discontinuity in ground truth, where a discontinuity is a step whose ratio of neighbouring depths exceeds a threshold τ. The selector is trained with binary cross-entropy averaged over the contour pixels only, since pixels off the contours are never candidates.

IV. Experiments

A. Setup

We evaluate on the 654-frame NYUv2 [12] test split, scene-disjoint from the training frames. The two transfer sets of Sec. IV-E are the 800-frame ScanNet [13] validation sample used by [3] and a 300-frame scene-level hold-out of Hypersim [14]. Ground truth is used with each dataset's own validity mask and depth range, and NYUv2 additionally with the Eigen crop. The metric is the scale-invariant boundary BF_1 of Depth Pro [5]. We report the relative change in BF_1 against each base model's own unsharpened prediction. Predictions are aligned to ground truth in their own space before scoring (log depth for Marigold V2, disparity for Lotus) and the alignment affects every variant equally.

Depth models. We evaluate two models that are as far apart as the family allows: Marigold V2, a 21.3B-parameter diffusion transformer fine-tuned with 4-bit QLoRA that emits affine-invariant log depth, and Lotus, an 868M-parameter UNet that emits disparity. Neither reports BF_1 on NYUv2, so we measure the baselines ourselves.

Choice of benchmarks. BF_1 compares neighbouring pixels, so it needs ground truth dense enough for both to be valid, and three of the five benchmarks these models are evaluated on do not provide it: KITTI [15] leaves 13.2% of adjacent pairs valid and ETH3D [16] 4.1%, while DIODE [17] is dense but its evaluation protocol masks the depth gradients this method operates on, removing 99.4% of indoor discontinuities. We therefore evaluate on NYUv2 and ScanNet, and additionally on Hypersim as a third transfer set.

Settings of the method. Barrier sharpening uses $r_b = 2$ and $r_f = 3$. The selector is a four-level U-Net with 1.9M parameters. On the training frames 6.066% of contour pixels lie within 1 px of a true step, so the retention is $\rho = 6\%$. The discontinuity threshold is $\tau = 10\%$, chosen on a validation split rather than on the test set: scoring BF_1 on the 141 frames of the 40 scenes held out of the 795 training frames, 10% gives $+12.74\%$ against $+8.66$, $+6.29$ and $+5.28\%$ for 15, 20 and 25%. Raising it thins the positives (from 6.07% of contour pixels to 2.13%) so the fixed budget is then spent mostly on negatives. Every hyperparameter of the method was set on the validation split; the one exception is a control, and Sec. IV-E states it there. The 795 NYUv2 training frames are split by scene into 654 for training and 141 for validation, and the best-validation epoch is used. Training takes 16 minutes and 3.5 GB of VRAM on one RTX 4090.

Implementation. The selector is trained with AdamW at a learning rate of 3×10^{-4}, weight decay 10^{-4} and a cosine schedule, for 40 epochs at batch size 4. Frames are used at their native 480×640 with no resizing and no augmentation: the band and the refill radius are defined in pixels, so rescaling would change what $r_b = 2$ means. Contours come from Segment Anything [11] with the ViT-H checkpoint in automatic mode, a 48×48 prompt grid and its default quality filter; "grid 16" and "grid 32" in Sec. IV-E vary that grid. A frame's contour set is the union of the one-pixel boundaries of the returned masks, not the union of the masks, which would fill the frame.

B. Why selection is needed

Applied at every contour that Segment Anything draws on NYUv2, barrier sharpening lowers Marigold V2's boundary BF_1 by 1.90% (Table I).

TABLE I
WHY POST-HOC SHARPENING FAILS ON MARIGOLD V2, AND WHAT FIXES IT. NYUV2, 654 FRAMES. ORACLE SELECTION USES GROUND TRUTH TO CHOOSE A SUBSET OF THE SEGMENTATION CONTOURS AND MOVES NONE OF THEM.

Contours sharpened	share kept	BF_1 change
all segmentation contours	100%	-1.90%
oracle selection, within 1 px of a step	6.07%	$+\mathbf{43.39}\%$

TABLE II

MAIN RESULT. FOUR SEEDS; ± IS ONE STANDARD DEVIATION. IN ABSOLUTE TERMS THE SELECTOR TAKES BF_1 FROM 0.0884 TO 0.0989 AND FROM 0.0693 TO 0.0886.

Base depth model	BF_1	no sel.	+ selector
Marigold V2, 21.3B, log	0.0884	-1.90%	$\mathbf{+11.92 \pm 0.90\%}$
Lotus, 868M, disparity	0.0693	$+10.31\%$	$\mathbf{+27.83 \pm 1.27\%}$

Measuring from each true depth discontinuity to the nearest contour gives a median of 1.00 px, with 66.9% within 1 px and 76.9% within 2 px. The contours are where the boundaries are, but 79.8% of the contour pixels have no depth step anywhere within an eight-pixel window; these are wallpaper patterns, floor seams and poster edges. Sharpening there introduces a discontinuity into a region that has none.

The 79.8% uses a generous eight-pixel window; the selection below uses a strict one-pixel tolerance. Of the 22,167 segmentation contour pixels in an average frame (10.6% of its valid pixels) 1,345, or 6.07%, lie within 1 px of a true discontinuity. An oracle that keeps exactly those, and leaves every kept contour where it is, moves the effect from -1.90% to $+43.39\%$ (Table I). The loss is therefore attributable to which contours are sharpened rather than to where they lie, and the selector of Sec. III-C learns that selection.

C. Main result

Table II gives the result on two depth models that share no architecture, no parameter scale and no output representation. Four selectors are trained, differing only in seed, all on Marigold V2's predictions. The same four are applied to Lotus with no retraining, adding only the disparity-to-log-depth conversion.

All four seeds clear $+10\%$ on Marigold V2, with a 95% confidence interval whose lower bound is $+10.49\%$. Pixel-wise accuracy does not regress either: AbsRel moves from 0.03881 to 0.03860 and δ_1 from 0.97969 to 0.98000, both slightly in the better direction.

Per-frame scoring shows the mean is not carried by a few frames: 484 of 651 frames improve (74.3%), with a median relative gain of $+13.15\%$ and first and third quartiles -0.19% and $+31.57\%$; a quarter of frames are therefore slightly worse. Three frames with a baseline BF_1 of zero are excluded, as their relative gain is undefined.

Fig. 2 shows an example frame, one of those with a large gain among the frames whose AbsRel is at or below the median. Over the whole frame, nothing outside the bands around the selected contours changes. In the middle row, a crop at the lower part of a chair, the step that the baseline spreads over several pixels becomes steep after sharpening. Comparing the depth discontinuities in the bottom row, along the vertical edge of the chair the pixels that the baseline left in the prediction only (red) or in ground truth only (grey) become, after sharpening, pixels that agree with ground truth (green).

TABLE III

SOFT EDGE ERROR, FOR ONE OF THE FOUR SEEDS OF TABLE II (THE ONE SCORING $+12.49\%$, SECOND BEST OF THE FOUR) APPLIED UNCHANGED TO BOTH MODELS. LOWER IS BETTER; PERCENTAGES ARE RELATIVE TO EACH MODEL'S OWN BASELINE. "GT CONTOURS" SHARPENS AT THE GROUND-TRUTH DISCONTINUITIES THEMSELVES AND IS NOT REACHABLE BY SELECTING AMONG SEGMENTATION CONTOURS; THE ORACLE SELECTION OF TABLE I IS A DIFFERENT AND MUCH SMALLER BOUND.

	BF_1	SEE_3	SEE_5	SEE_7
Marigold V2				
baseline	0.0884	0.1148	0.0882	0.0758
+ selector	$+12.49\%$	-3.34%	-2.89%	-2.39%
relocated	-0.50%	$+0.32\%$	$+0.46\%$	$+0.52\%$
GT contours	$+209.8\%$	-11.44%	-5.09%	-2.91%
Lotus				
baseline	0.0693	0.1403	0.1147	0.1014
+ selector	$+28.18\%$	-5.18%	-5.54%	-5.48%
relocated	-0.57%	$+0.20\%$	$+0.23\%$	$+0.24\%$
GT contours	$+279.0\%$	-12.96%	-9.81%	-8.73%

D. Evaluation with Soft Edge Error

An improvement visible in one metric may be an improvement in that metric's definition. We therefore also report Soft Edge Error [6], the boundary metric Marigold V2 adopts, transcribed from its implementation. SEE_k averages, over ground-truth boundary pixels, the smallest absolute error between the prediction and any ground-truth depth in a $k \times k$ neighbourhood; it forgives a displaced boundary and punishes a blurred one. Unlike BF_1 it is an absolute error in depth units, so it is not scale-invariant and the prediction must be aligned first. Lower is better.

The two metrics agree on every variant of both models: the selector improves both, the relocated control worsens both, and sharpening at GT contours is best on both. The improvement is therefore not an artefact of BF_1's definition.

E. Analysis

This subsection analyses where the improvement comes from and how far it holds. We first compare against training-free controls to confirm that the gain comes from the learned selection. We then measure the contribution of the segmentation model and test how the selector generalises to other datasets. Finally, we report the cost.

Controls. The gain could come from the sharpening rather than from the learning, or from simply being sparse. Table IV measures six training-free alternatives, each on the base model's own predictions.

The edge-preserving filters lose on both models, monotonically in the radius, from -4.75% and -7.15% at the gentlest setting to -50.14% and -45.30%. A filter can only smooth, and a smoothed boundary is what BF_1 counts against.

The hand-crafted control thresholds a local descriptor of the predicted depth's log-gradient. We select its feature by the final metric on the test split rather than by ranking quality on held-out data. That gives the control an advantage the method does not have, which makes the comparison conservative.

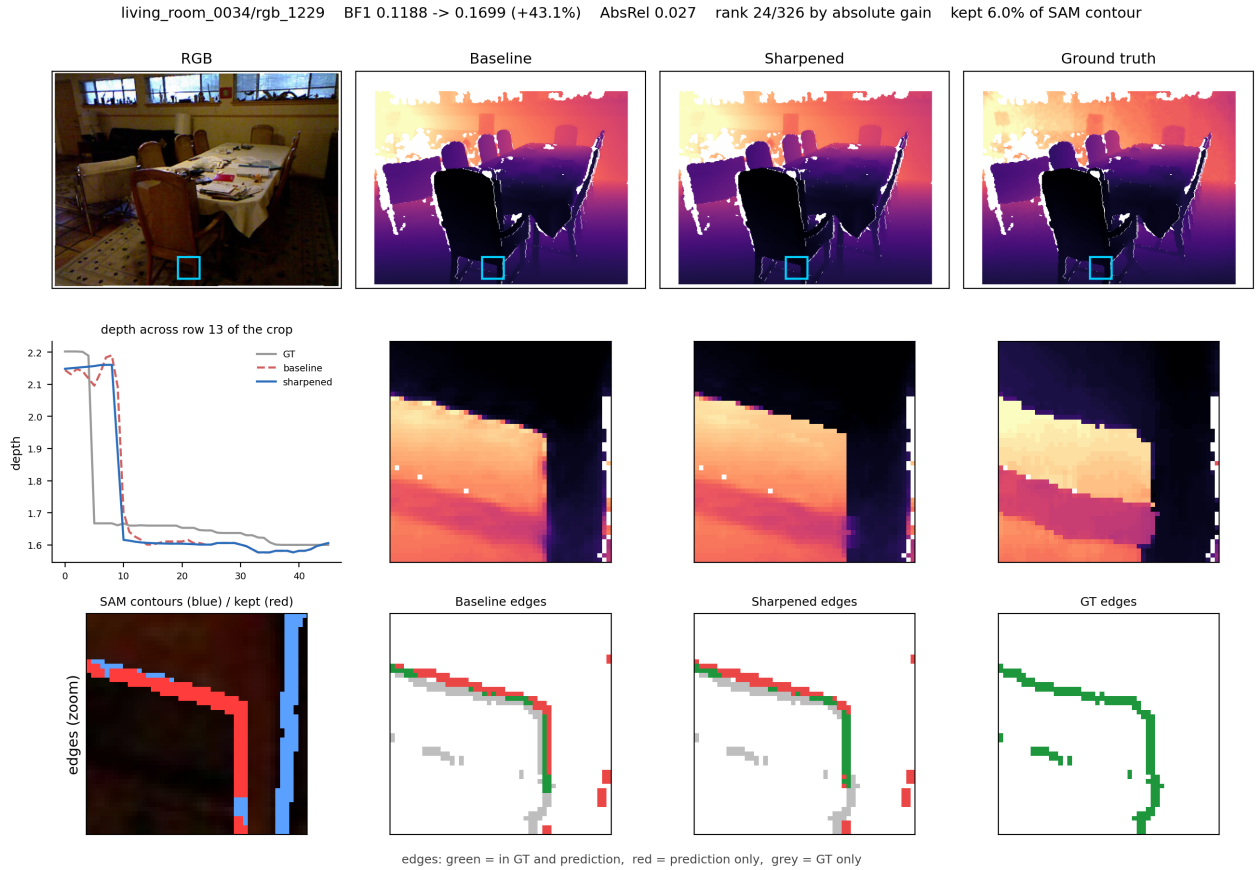


Fig. 2. Example of the method on a living-room frame. Top: the whole frame. Middle: the 48 px crop marked by the cyan box, with a depth profile across it; the three maps share a colour scale taken from the crop. Bottom: the segmentation contours (blue) and those the selector kept (red), followed by the depth discontinuities of each map; green is in both ground truth and prediction, red in the prediction only, grey in ground truth only.

TABLE IV

TRAINING-FREE CONTROLS, MEASURED SEPARATELY ON EACH BASE MODEL. BF_1 CHANGE. THE RELOCATED ROW IS THE MEAN OF THREE INDEPENDENT RELOCATIONS; THE TWO RANDOM ROWS ARE MEANS OF THREE DRAWS, EACH SPENDING THE SELECTOR'S OWN PER-FRAME BUDGET. THE FILTER ROW IS THE BEST OF A GUIDED AND A JOINT BILATERAL FILTER OVER FOUR RADII, JOINT BILATERAL AT RADIUS 1 ON BOTH.

Method	Marigold V2	Lotus
none (all contours)	-1.90%	$+10.31\%$
relocated (same sparsity)	-0.50%	-0.57%
random pixels	$-24.96 \pm 0.53\%$	$-22.11 \pm 1.17\%$
random blocks	$-1.04 \pm 0.15\%$	$-0.03 \pm 0.36\%$
best hand-crafted threshold	$+1.25\%$	$+10.46\%$
joint bilateral filter, r=1	-4.75%	-7.15%
trained selector learning contributes	$\mathbf{+11.92}\%$ **10.7** pt	$\mathbf{+27.83}\%$ **17.4** pt

Random-pixel selection is close on the two models (-24.96% and -22.11%), which is what a control that measures the repair operation rather than the depth model should do. Random blocks, which keeps the contour contiguous, costs almost nothing on either and nothing at all on Lotus: scattering the selection is most of what makes the pixel control so destructive, and choosing badly on its own incurs almost no cost.

Contribution of the segmentation model. Removing the segmentation model entirely, so that the selector must find boundaries itself from RGB and depth and every valid pixel is a candidate, gives $+7.30\%$. Both halves of this ablation come from a single seed, so the figure to compare against is that seed's $+12.49\%$: the segmentation is worth 5.19 points, 42% of that run's gain. This contribution arises on the recall side rather than the precision side. On segmentation contours the true steps number 1,345 pixels per frame and the budget of 1,330 can cover 98.9% of them; counting over the whole valid image there are 2,941, and the same budget caps recall at 45.2%. The segmentation earns its contribution by *shrinking the target set*.

Cross-dataset generalisation. The selector transfers across depth models but not across datasets. Table V applies the NYUv2-trained selector to ScanNet and Hypersim, using Marigold V2 throughout.

On neither held-out dataset does the selector beat using every contour. What fails is the transfer, not the datasets. A selector trained on 10,000 Hypersim frames reaches $+5.05\%$ on Hypersim itself, against $+2.28\%$ for no selection, at 88.1%

TABLE V
GENERALISATION. THE SELECTOR WAS TRAINED ON NYUV2; THE DEPTH MODEL IS MARIGOLD V2 IN EVERY ROW.

Evaluation data	no selection	+ selector
NYUv2 (in-domain)	-1.90%	$\mathbf{+11.92\%}$
ScanNet	$+3.26\%$	$+2.81\%$
Hypersim	$+2.28\%$	-4.64%

TABLE VI
PER-FRAME RUNNING TIME AND THE IMPROVEMENT OBTAINED. GRIDS 16 AND 48 ARE MEANS OVER THE FOUR SEEDS OF TABLE II WITH THEIR STANDARD DEVIATION; THE OTHER TWO ROWS ARE THE SINGLE SEED SCORING $+12.49\%$ THERE. THE SEGMENTATION POINT GRID IS THE ONLY SETTING WITH A SUBSTANTIAL EFFECT.

Configuration	ms/frame	BF_1 change	ms/BF_1 pt
depth model only	495	—	—
+ selector, no seg.	525	$+7.30\%$	**72**
+ selector, grid 16	**1,059**	$\mathbf{+11.02 \pm 0.64\%}$	**96**
+ selector, grid 32	1,996	$+11.17\%$	179
+ selector, grid 48	3,128	$+11.92 \pm 0.90\%$	262

precision on a 22.8% base rate; the same checkpoint reaches -2.23% on NYUv2 and $+0.39\%$ on ScanNet. Hypersim is not saturated either (sharpening at its own GT contours gives $+39.49\%$) so the method works in its domain as it does in NYUv2's, and stops working only when the domain changes between training and use.

Three further remedies failed: dropping the RGB channels, sweeping the retention from 2% to 20% on ScanNet, and training on an equal mixture of 795 NYUv2 and 795 Hypersim frames. Data volume does not account for the gap either. Cutting NYUv2 to 393 frames costs 2.2 points of gain ($+11.92\%$ to $+9.72\%$, two seeds), against the 9.1 points that separate in-domain from ScanNet.

Cost. Table VI gives wall-clock time per frame on one RTX 4090, measured after warm-up on real frames.

The selector costs 12.0 ms and the repair 18.3 ms, so what this work adds is 30.3 ms, 6% of the depth model's own 495 ms. The segmentation dominates everything else, at 534 ms to 2,603 ms depending on its point grid.

Coarseness of the point grid. Grid 16 reaches 92% of grid 48's gain for 20% of the segmentation cost. The 0.90-point gap equals the standard deviation across seeds, so a single pair of runs cannot settle the ordering; paired by seed, however, grid 48 is ahead in all four, by 0.90 ± 0.29 points (paired t-test, $p = 0.009$). The difference is consistent and small.

V. CONCLUSION

This paper proposed a post-processing method that improves the object boundaries blurred by diffusion-based monocular depth estimators without modifying the depth model. It fixes barrier sharpening, which restores depth steps at the contours of a segmentation model, and uses a small U-Net to select only the contours it is applied to.

The experiments yield three findings. First, sharpening at every contour lowers boundary accuracy, and the cause is not where the contours lie but that contours carrying no depth step are sharpened too. Second, learning which contours to sharpen improves the boundary BF_1 of Marigold V2 on NYUv2 by 11.9%, and the same selector applied to Lotus without retraining improves it by 27.8%; training-free controls and classical filters do not achieve this, and Soft Edge Error confirms it. Third, the selector transfers across depth models but not across datasets.

Limitations. The oracle selection of Sec. IV-B reaches $+43.39\%$ against our $+11.92\%$; the gap lies in the precision of the selection, 50.2% against 100%. For AbsRel and δ_1 we claim only that they do not regress. Supervision and BF_1 both refer to ground-truth depth steps; ground truth never enters inference, but the shared definition may bear on how the result is read. Making the selection independent of the training domain is the main open problem.

ACKNOWLEDGMENT

Claude Code (Anthropic) was used to implement the code and to write the initial draft of this paper. The author subsequently reviewed the whole and takes responsibility for its content.

REFERENCES

[1] B. Ke, A. Obukhov, S. Huang, N. Metzger, R. C. Daudt, and K. Schindler, "Repurposing diffusion-based image generators for monocular depth estimation," in *CVPR*, 2024.

[2] J. He, H. Li, W. Yin et al., "Lotus: Diffusion-based visual foundation model for high-quality dense prediction," in *ICLR*, 2025.

[3] I. Pavlovic, T. Wandel, A. Obukhov, L. Bartolomei, A. Davydov, F. Tosi, M. Poggi, S. Süsstrunk, and D. Dai, "Marigold V2: Revisiting diffusion transformers for monocular depth estimation," arXiv:2609.08084, 2026.

[4] G. M. Garcia, K. Knaebel, C. Schmidt, D. de Geus, A. Hermans, and B. Leibe, "Fine-tuning image-conditional diffusion models is easier than you think," in *WACV*, 2025.

[5] A. Bochkovskii et al., "Depth Pro: Sharp monocular metric depth in less than a second," in *ICLR*, 2025.

[6] F. Tosi, Y. Liao, C. Schmitt, and A. Geiger, "SMD-Nets: Stereo mixture density networks," in *CVPR*, 2021.

[7] M. Ramamonjisoa, Y. Du, and V. Lepetit, "Predicting sharp and accurate occlusion boundaries in monocular depth estimation using displacement fields," in *CVPR*, 2020.

[8] M. Ramamonjisoa and V. Lepetit, "SharpNet: Fast and accurate recovery of occluding contours in monocular depth estimation," in *ICCV Workshops*, 2019.

[9] K.-S. Peng, G. Ditzler, and J. Rozenblit, "Edge-guided occlusion fading reduction for a light-weighted self-supervised monocular depth estimation," arXiv:1911.11705, 2019.

[10] A. Muthukkumar, "Bayesian monocular depth refinement via neural radiance fields," in *Proc. IEEE 8th Int. Conf. Algorithms, Computing and Artificial Intelligence (ACAI)*, 2025, pp. 488–492.

[11] A. Kirillov et al., "Segment Anything," in *ICCV*, 2023.

[12] N. Silberman, D. Hoiem, P. Kohli, and R. Fergus, "Indoor segmentation and support inference from RGBD images," in *ECCV*, 2012.

[13] A. Dai, A. X. Chang, M. Savva, M. Halber, T. Funkhouser, and M. Nießner, "ScanNet: Richly-annotated 3D reconstructions of indoor scenes," in *CVPR*, 2017.

[14] M. Roberts et al., "Hypersim: A photorealistic synthetic dataset for holistic indoor scene understanding," in *ICCV*, 2021.

[15] A. Geiger, P. Lenz, and R. Urtasun, "Are we ready for autonomous driving? The KITTI vision benchmark suite," in *CVPR*, 2012.

[16] T. Schöps, J. L. Schönberger, S. Galliani, T. Sattler, K. Schindler, M. Pollefeys, and A. Geiger, "A multi-view stereo benchmark with high-resolution images and multi-camera videos," in *CVPR*, 2017.

[17] I. Vasiljevic et al., "DIODE: A dense indoor and outdoor depth dataset," arXiv:1908.00463, 2019.